



022 E1004

S.V. National Institute of Technology, Surat
Department of Electronics Engineering
MID Semester Exam, September-2024
B. Tech. (EC) V Semester
Communication Networks (EC307)

Date: 27/09/2024 & Time: 11:00 AM to 12:30 PM

Maximum Marks: 30

Duration: 1:30 Hours

Instructions:

1. [CO] indicates course outcome.
2. All the questions are compulsory.
3. Marks distribution for each question is clearly provided in the 'Marks' column.

Q.1	Write the answers in a few words.	Marks	CO
a	Answer the following questions based upon the figure given below. a. This frame corresponds to which protocol? (1 Mark) b. What is the type of this frame? (1 Mark) c. In which phase, this frame is used? (1 Mark) d. Name one method that can be used for FCS? (1 Mark) e. What is 'B'? (1 Mark)	5	[CO2]
	F i a g B Control 1111100 Mg. data F C S F i a g		
b	Explain Latency and its components.	3	[CO1]
c	Draw the flow chart of ALOHA protocol and explain it.	3	[CO2]
d	Write the names of each layer and the works associated with each layer.	2	[CO1]
e	Explain Go-back-N protocol with proper timing diagrams and sliding window diagrams.	5	[CO2]
f	Explain the structure of Optical fiber cable.	2	[CO1]
Q.2	Write the short answers to the following Questions.		
a	Consider a pair of sender and receiver using Selective repeat ARQ algorithm at Data Link layer. The pair is using Modulo – 4 numbering for providing the sequence numbers to the frames. The size of sender and receiver windows is 3. Draw the clear timing diagram showing the sliding sender and receiver windows considering the following events. A. First frame is sent by the sender. It is successfully received by the receiver. Receiver sends the ACK and it is also successfully received by the sender. B. Second frame is sent by the sender. It is received by the receiver with some error. C. Second frame is resent by the sender. It is successfully received by the receiver. Receiver sends the ACK but it is not received successfully by the sender. D. Third frame is sent by the sender. It is successfully received by the receiver. Receiver sends the ACK but it is not received successfully by the sender.	5	[CO4]

	E. Fourth frame is sent by the sender. It is successfully received by the receiver. Receiver sends the ACK but it is not received successfully by the sender. F. After step E, Draw the next frame that will be transmitted from the sender and will be received by the receiver. (In this step, assume that the receiver will successfully receive the frame and the ACK will be successfully received by the sender). Comment whether this frame is correctly received or not with reason. (Clearly show every step including sliding window etc.)		
	h In case of CDMA, design a Walsh table for a network having 6 Nodes (Let the names of the nodes be A, B, C, D, E and F) (2 Marks). Assign one code to each node. Node A is transmitting bit '1', B is transmitting bit '0' and C is transmitting bit '1' and Node D is receiving. Show how D can detect the bits transmitted by A, B and C. (1+1+1 Marks) (Only theoretical explanation is not sufficient. Draw the appropriate codes/data/matrix/graph.)	5	[CO1]
